LAPORAN WORKSHOP DEVELOPER OPERATIONAL

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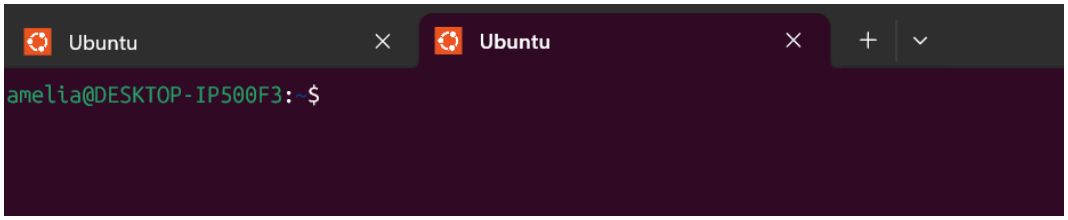
HASIL DAN PEMBAHASAN

Pada gambar dibawah ini, saya menggunakan 2 terminal Ubuntu untuk mensimulasikan dua node dalam Docker Swarm.

1. Ubuntu pertama digunakan sebagai manager, yaitu untuk mengatur dan mengontrol sistem
2. Ubuntu kedua digunakan sebagai worker, yaitu untuk menjalankan perintah dari manager

Penggunaan 2 Ubuntu ini bertujuan untuk mensimulasikan sistem terdistribusi meskipun hanya menggunakan satu perangkat.



Berikut merupakan langkah-langkah yang dilakukan dalam implementasi Docker Swarm mulai dari menjalankan Docker hingga menjalankan service.

1. **Menjalankan Docker**

   Pada tahap ini dilakukan pengecekan dan menjalankan Docker untuk memastikan Docker siap digunakan.

```
amelia@DESKTOP-IP500F3:~$ docker --version
Docker version 29.2.1, build a5c7197
amelia@DESKTOP-IP500F3:~$ sudo systemctl start docker
[sudo] password for amelia:
amelia@DESKTOP-IP500F3:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

2. **Mengecek IP Address**

   Digunakan untuk mengetahui alamat IP yang akan digunakan saat inisialisasi Docker Swarm.

```
amelia@DESKTOP-IP500F3:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:10:e1:15 brd ff:ff:ff:ff:ff:ff
    inet 172.22.97.143/20 brd 172.22.111.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe10:e115/64 scope link
       valid_lft forever preferred_lft forever
```

3. **Inisialisasi Docker Swarm (Manager)**

   Mengaktifkan mode swarm dan menjadikan node sebagai manager.

```
amelia@DESKTOP-IP500F3: $ docker swarm init --advertise-addr 172.22.97.143
Swarm initialized: current node (cg72nherygsixhqykeegi6e87) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-5vf4tzyv018f0n4ow60d9k6o716dybi3ibrcyq8y8b69xkuwfx-66kneoifauyoy3lolu13xpn5z 172.22.97.143:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

4. **Membuat Worker Node**

Membuat node worker menggunakan container sebagai simulasi node kedua.

```
amelia@DESKTOP-IP500F3:~$ docker run -dit --name worker1 --privileged docker:dind
2c6311af1d81479dc2bb375b341b02c2925aa465ee41e2c1f591b03d67ad809e
```

```
amelia@DESKTOP-IP500F3:~$ docker node ls
ID                            HOSTNAME          STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VER
SION
wp5496xaj62tk6h7vh2z5keoe     2c6311af1d81      Ready     Active                          29.3.0
cg72nherygsixhqykeegi6e87 *   DESKTOP-IP500F3   Ready     Active         Leader           29.2.1
```

5. **Join Worker ke Swarm**

   Worker bergabung ke dalam cluster menggunakan token dari manager.

```
amelia@DESKTOP-IP500F3:~$ docker exec -it worker1 sh
/ # docker swarm join --token SWMTKN-1-5vf4tzyv018f0n4ow60d9k6o716dybi3ibrcyq8y8b69xkuwfx-
66kneoifauyoy3lolu13xpn5z 172.22.97.143:2377
This node joined a swarm as a worker.
```

6. **Verifikasi Node**

   Digunakan untuk memastikan manager dan worker sudah terhubung.

```
amelia@DESKTOP-IP500F3:~$ docker node ls
ID                            HOSTNAME          STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VER
SION
wp5496xaj62tk6h7vh2z5keoe     2c6311af1d81      Ready     Active                          29.3.0
cg72nherygsixhqykeegi6e87 *   DESKTOP-IP500F3   Ready     Active         Leader           29.2.1
```

7. **Membuat Service NGINX**

   Menjalankan service web server dengan 2 replica untuk simulasi load balancing.

   docker service create --name web --replicas 2 -p 8085:80 nginx

8. **Cek Service**

   Memastikan service berjalan dengan jumlah replica sesuai.

```
amelia@DESKTOP-IP500F3:~$ docker service ls
ID             NAME      MODE         REPLICAS   IMAGE          PORTS
tymyvn8ti1fo   web       replicated   2/2        nginx:latest   *:8085->80/tcp
```

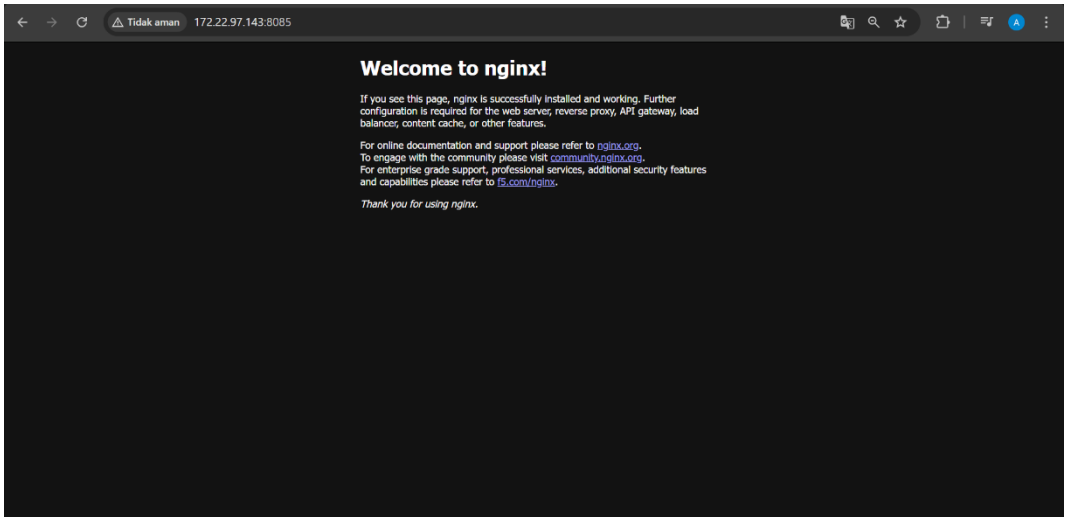
9. **Cek Detail Service**

   Melihat status container yang berjalan pada setiap node.

```
amelia@DESKTOP-IP500F3:~$ docker service ps web
ID             NAME        IMAGE          NODE              DESIRED STATE   CURRENT STATE
  ERROR     PORTS
jwmfw5hucsok   web.1       nginx:latest   DESKTOP-IP500F3   Running         Running 11 minutes ago

qe0qg176sojn    \_ web.1   nginx:latest   DESKTOP-IP500F3   Shutdown        Shutdown 11 minutes ago

907nyfcnivem    \_ web.1   nginx:latest   DESKTOP-IP500F3   Shutdown        Shutdown 13 minutes ago

nwbxt772zk21    \_ web.1   nginx:latest   2c6311af1d81      Shutdown        Shutdown 14 minutes ago

mo3aunvx3rv2   web.2       nginx:latest   DESKTOP-IP500F3   Running         Running 11 minutes ago

tk8v41jibfhr    \_ web.2   nginx:latest   DESKTOP-IP500F3   Shutdown        Shutdown 11 minutes ago

rpauierifqw2    \_ web.2   nginx:latest   DESKTOP-IP500F3   Shutdown        Shutdown 13 minutes ago

sj2eehkhas51    \_ web.2   nginx:latest   DESKTOP-IP500F3   Shutdown        Shutdown 13 minutes ago
```

10. **Akses di Browser**

    Menguji apakah service berhasil dijalankan melalui browser.


Tidak aman
172.22.97.143:8085
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.